

Yutong Wang

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EDUCATION

Brown University, GPA: 4.00/4.00

Providence, RI | **Expected Graduation May 2025**

B.S. Computer Science, B. A. International and Public Affairs

Courses: Software Engineering, Computer Systems, Software Security and Exploitation, Deep Learning

Clubs and Activities: Brown Motion Pictures, Brown Journal of World Affairs

WORK AND RESEARCH EXPERIENCE

Horizon Robotics (Advanced driver assistance systems (ADAS) integrated solutions provider)

Machine Learning Infrastructure - Software Engineering intern

Shanghai, China | Jun 2023 – Aug 2023

- Contributed to the development of Horizon Data Flow in **Python**, an internal data management software for machine learning data preprocessing. Specifically optimized three data annotation algorithms for autonomous vehicle perception data, achieving an accuracy of about 90% and doubled efficiency
- Maintained 5 **MongoDB** and **Apache Doris** databases for autonomous vehicle perception data, and optimized data structure that resulted in improved interpretability and 4 times faster query response speed
- Produced 2 dynamic data visualization dashboards using **Apache Superset** with labeled datasets, and developed customizable filters to perception and trajectory planning teams for meta-analysis of projects
- Designed an internal tool to convert perception data to various formats to be visualized on **Foxglove Studio**, resulting in a time-saving of over 60% compared to the previous method, benefiting more than 300 colleagues

Automatic Coordination of Teams Lab (Research focus on coordinated multi-robot systems)

Research Assistant

Brown University | Sep 2022 – Present

- Studying motion coordination of drones and ground robots operating in proximity with **ROS2**, Crazyflie platform, and Vicon motion tracking system, optimizing obstacle detection and avoidance to improve navigation
- Designing the syllabus and projects for an intermediate-level interdisciplinary Robots and Art Course (CSCI 1952Z) scheduled for Spring 2024, anticipating an enrollment of approximately 40 students
- Developing a user-friendly translation of complex drone commands with Google's **Blockly** library

SafeMode Mobility (Startup focusing on driver incentive and engagement platform for commercial fleets)

Data Analyst Intern

Shanghai, China & Tel Aviv, Israel | Jun 2022 – Sep 2022

- Conducted market research and created content for email marketing campaigns using Apollo, performed A/B testing for content improvement, and increased customer base by about 30%
- Interviewed existing customers, wrote and edited 6 case studies for distribution online and at conferences
- Delegated to the 2022 Home Delivery World conference in Philadelphia, PA, and assisted the company founder in presentations to more than 200 prospective clients

PROJECTS

Animal Research Compliance Inspection Project | *TypeScript* | Jan 2023 – Present

- Developing a **Next.js** application to streamline animal research compliance inspections, enabling users to create semiannual inspection plans, log deficiencies during inspections, and automate report generation. This prototype is able to reduce inspection time by half and has garnered recognition from 50 universities
- Integrated an email reminder system using SendGrid API with dynamic templates, enhancing communication efficiency between the inspection team and laboratories. Creating front-end interfaces for inspection admins, inspectors, and lab PI.

TaskMaster – Task Management App | *Java, TypeScript* | Mar 2023 – May 2023

- Designed and developed a **React** application with a **Spark Java** backend for task organization, utilizing the earliest deadline first (EDF) algorithm to provide optimized time management recommendations
- Integrated the **Google Calendar API** to retrieve spare time data and post time management recommendations, leveraging **Google Firebase** to store data efficiently

Music Generation with WaveNet | *Python* | Dec 2022

- Created an object-oriented API of deep learning model WaveNet using **Python TensorFlow** for music generation, which accepted .midi (encoded on musical notes) and .wav (encoded on waves) as inputs

SKILLS

Programming languages: Python, Java, C, SQL, HTML, CSS, TypeScript, Go

Frameworks, Tools and Libraries: Git, Bash, React, Node.js, Flask, Spark, TensorFlow, MongoDB, CI / CD

Technical Skills: Microsoft Excel, Adobe Premiere Pro, Figma